

DBSCAN Clustering Implementation

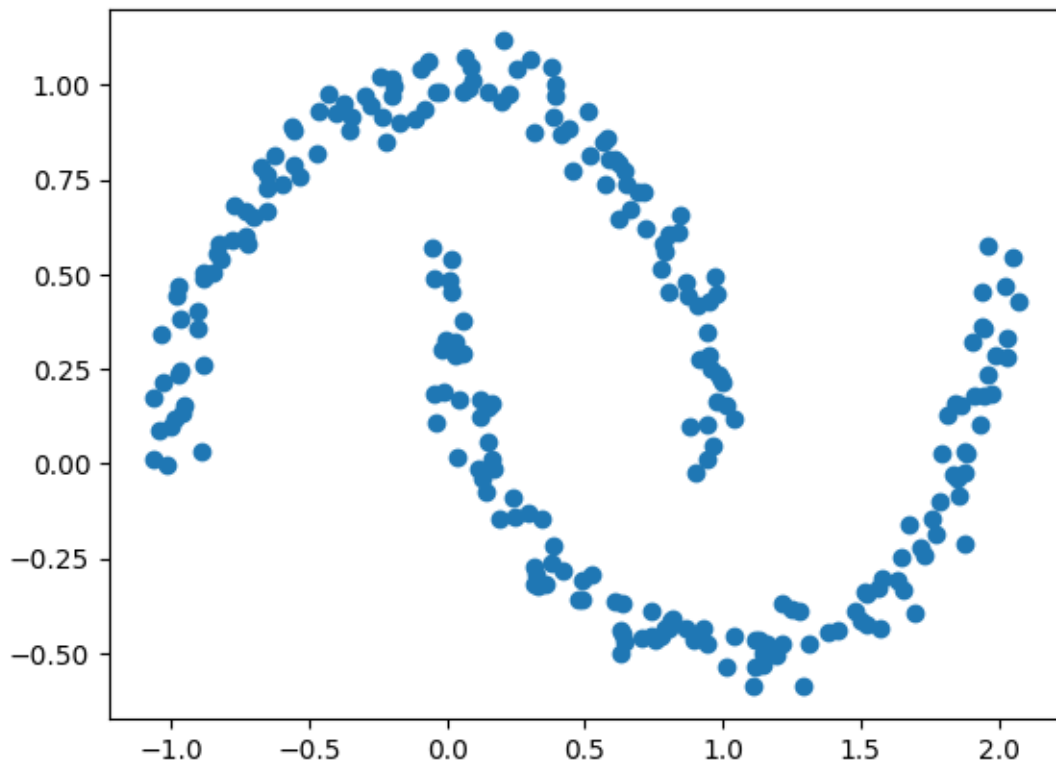
September 22, 2025

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
from sklearn.cluster import DBSCAN
from sklearn.datasets import make_moons
```

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[2]: X, y = make_moons(n_samples=250, noise=0.05)
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```
[3]: plt.scatter(X[:,0], X[:,1])
```

```
[3]: <matplotlib.collections.PathCollection at 0x20b88770fa0>
```



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[4]: from sklearn.preprocessing import StandardScaler  
     scaler = StandardScaler()
```

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[5]: X_scaled = scaler.fit_transform(X)
```

```
[6]: dbscan = DBSCAN(eps=0.3)  
     dbscan.fit(X_scaled)
```

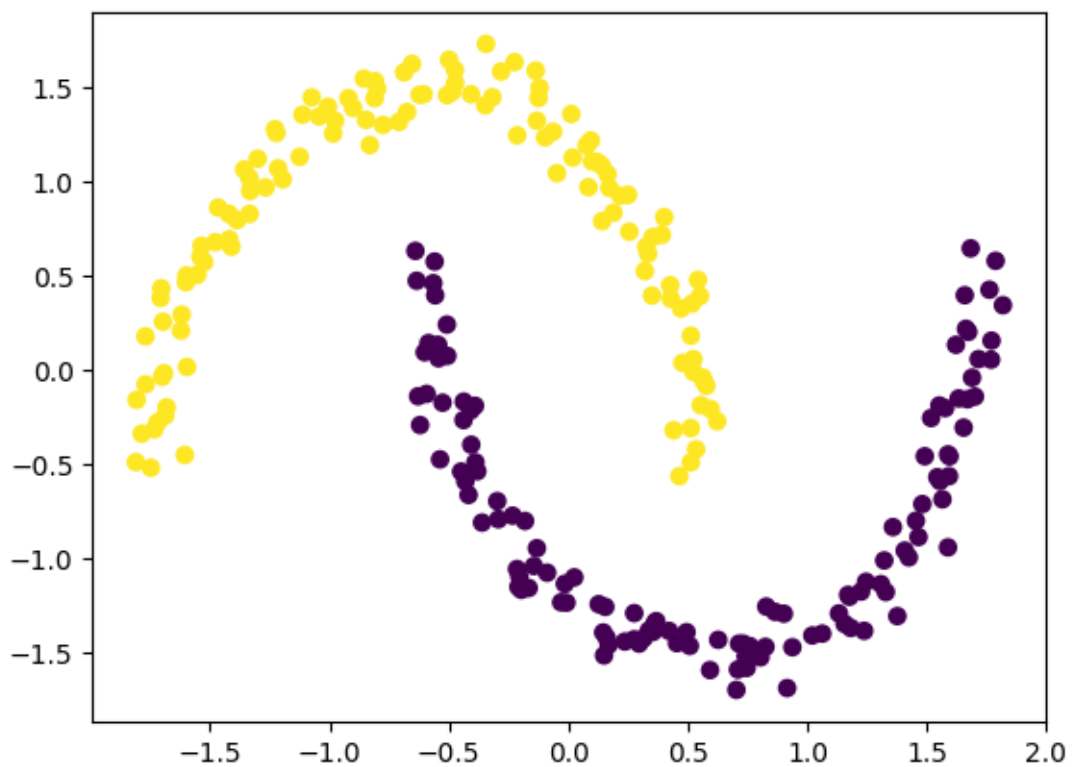
```
[6]: DBSCAN(eps=0.3)
```

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[7]: dbscan.labels_
```

```
[7]: array([0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 1, 0,  
          0, 1, 1, 1, 1, 1, 0, 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1,  
          0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1,  
          0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 0, 0,  
          1, 1, 1, 1, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
          1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1, 0, 0,  
          0, 1, 1, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0,  
          1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0,  
          1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1,  
          1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, 0,  
          1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0,  
          0, 0, 0, 1, 1, 0, 0, 1], dtype=int64)
```

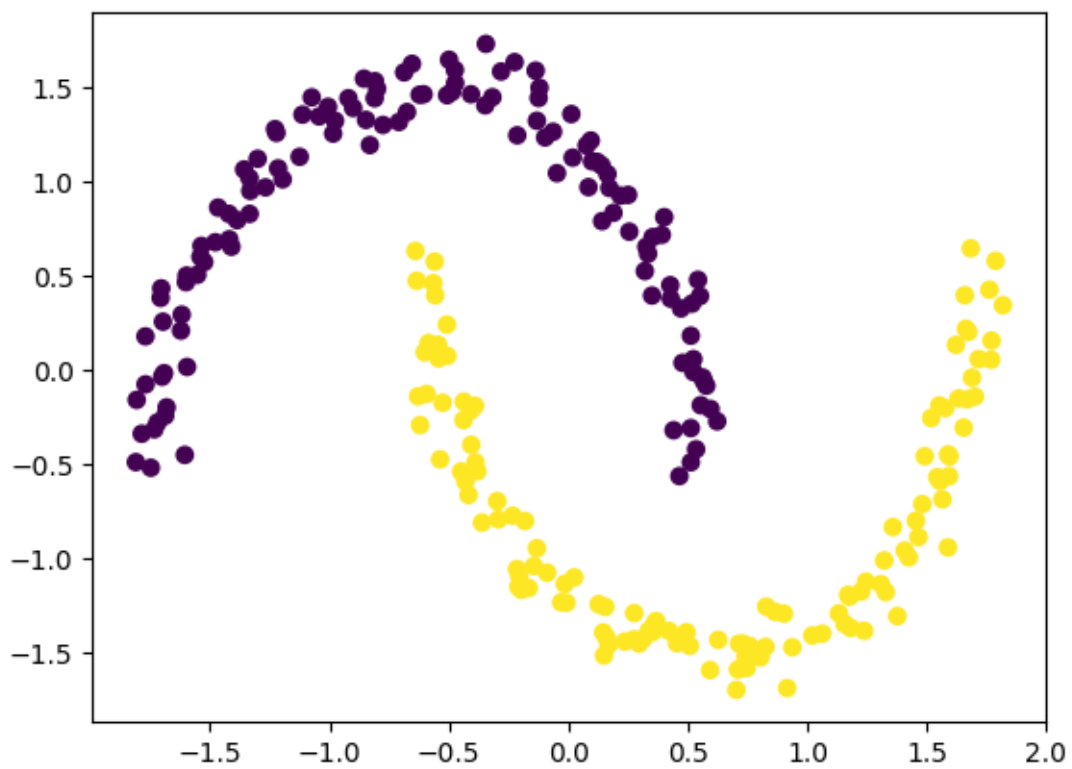
```
[8]: plt.scatter(X_scaled[:,0], X_scaled[:,1], c = dbscan.labels_)
```

```
[8]: <matplotlib.collections.PathCollection at 0x20b8aa2a560>
```



```
[9]: plt.scatter(X_scaled[:,0], X_scaled[:,1], c = y)
```

```
[9]: <matplotlib.collections.PathCollection at 0x20b8aad0c10>
```



[]: